LAST NAME: _____

Practice Quiz 07B — MATH 2055 — Fall 2026

Practice for Quiz 07

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Let $A = \{1, 2, 3\}$. Construct two nonempty transitive relations R and S on A such that $R \cup S$ is not transitive. List their ordered pairs, justify that each relation is transitive, and exhibit a failure of transitivity in their union.